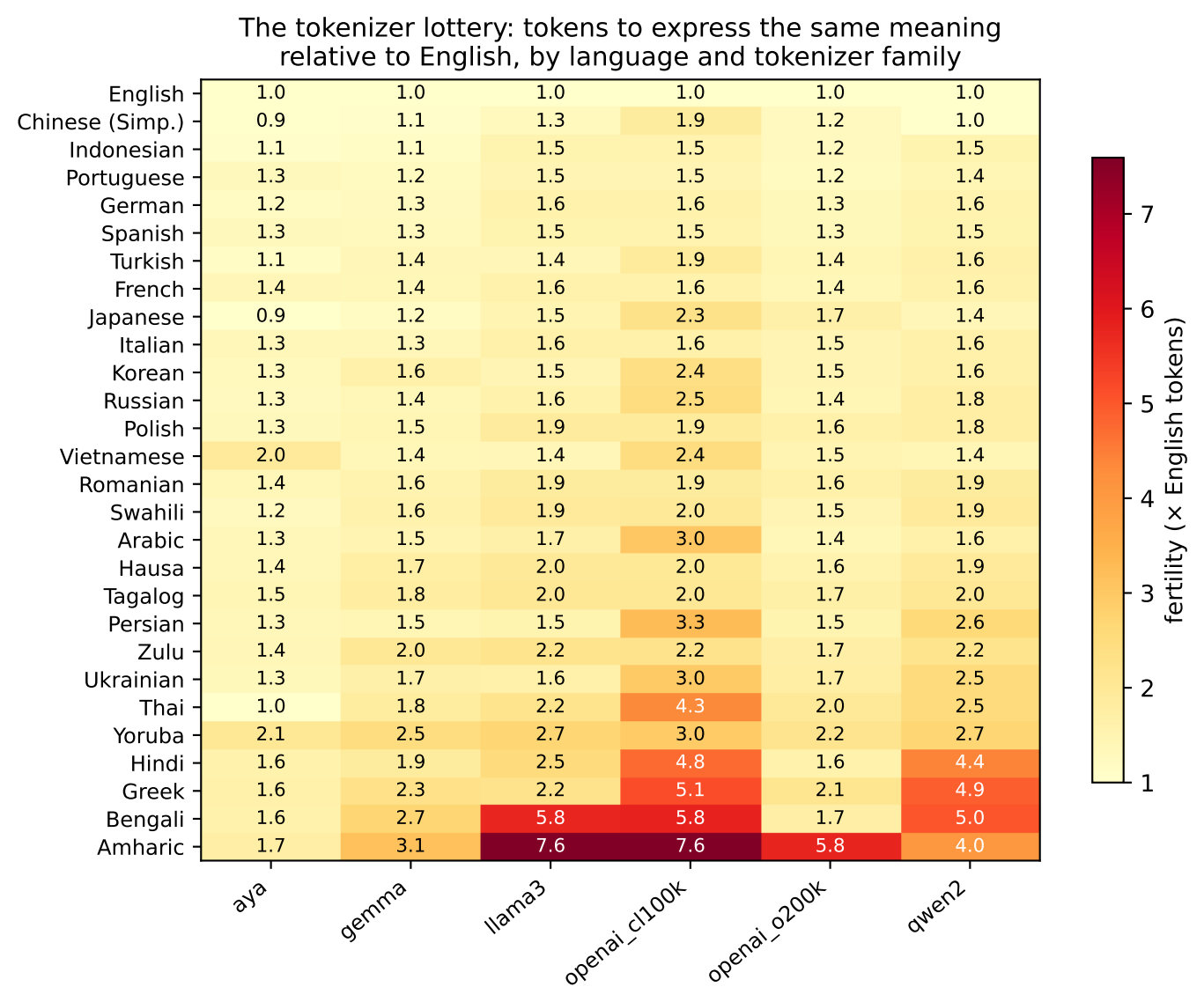
The tokenizer lottery: tokens to express the same meaning relative to English, by language and tokenizer family
English 1.0 1.0 1.0 1.0 1.0 1.0
Chinese (Simp.) 0.9 1.1 1.3 1.9 1.2 1.0
Indonesian 1.1 1.1 1.5 1.5 1.2 1.5
Portuguese 1.3 1.2 1.5 1.5 1.2 1.4
German 1.2 1.3 1.6 1.6 1.3 1.6
Spanish 1.3 1.3 1.5 1.5 1.3 1.5
Turkish 1.1 1.4 1.4 1.9 1.4 1.6
French 1.4 1.4 1.6 1.6 1.4 1.6
Japanese 0.9 1.2 1.5 2.3 1.7 1.4
Italian 1.3 1.3 1.6 1.6 1.5 1.6
Korean 1.3 1.6 1.5 2.4 1.5 1.6
Russian 1.3 1.4 1.6 2.5 1.4 1.8
Polish 1.3 1.5 1.9 1.9 1.6 1.8
Vietnamese 2.0 1.4 1.4 2.4 1.5 1.4
Romanian 1.4 1.6 1.9 1.9 1.6 1.9
Swahili 1.2 1.6 1.9 2.0 1.5 1.9
Arabic 1.3 1.5 1.7 3.0 1.4 1.6
Hausa 1.4 1.7 2.0 2.0 1.6 1.9
Tagalog 1.5 1.8 2.0 2.0 1.7 2.0
Persian 1.3 1.5 1.5 3.3 1.5 2.6
Zulu 1.4 2.0 2.2 2.2 1.7 2.2
Ukrainian 1.3 1.7 1.6 3.0 1.7 2.5
Thai 1.0 1.8 2.2 4.3 2.0 2.5
Yoruba 2.1 2.5 2.7 3.0 2.2 2.7
Hindi 1.6 1.9 2.5 4.8 1.6 4.4
Greek 1.6 2.3 2.2 5.1 2.1 4.9
Bengali 1.6 2.7 5.8 5.8 1.7 5.0
Amharic 1.7 3.1 7.6 7.6 5.8 4.0
aya
gemma
llama3
openai_cl100k
openai_o200k
qwen2
fertility (× English tokens)
1
2
3
4
5
6
7
FLORES-200 devtest · source Muennighoff/flores200 · measured 2026-07-06